

# Modified MAT135 Assignment 6 (Written Component)

Due 27 October 2024, at 9pm

## Submitting the Assignment

**Due date:** Sunday, 27 October 2024, at 9pm.

**Where to submit:** On Crowdmark. Check your email for an invitation to Crowdmark.

**What to do:** Write solutions to the problems on blank or lined sheets of paper, or write them digitally. Start each question on a new page. Then photograph, scan, or export a PDF of your solutions. See the [Written Assignments](#) page on Quercus for details.

**Uploading:** Click on the link in the email from Crowdmark to submit your solutions. Plan on giving yourself some extra time to upload your solutions the first time.

## More details

0. **What's this new version?** There's only one change, to the definition the Background section. It makes Q1(a) into the quick warm-up question we intended it to be. **The change is in red below, and only affects the answer to Q1(a).**
1. **Deadline:** The deadline to submit this assignment is strict, *to the second*. Assignments that are even a few seconds late will normally receive a grade of 0. Technical issues are not a valid reason to be late, so you are taking a risk if you leave uploading your solutions to the last minute. **Please give yourself lots of time to upload your assignment!**
2. **Purpose and feedback:** The purpose of the written assignments is to give you some practice in thinking about and writing solutions to mathematical problems, without any time pressure. You will receive feedback on your writing and on your solutions. You are encouraged to take this opportunity to carefully write your solutions and think about how to best present your reasoning behind them.
3. **Writing solutions:** *Explain all your work*, show your steps as well as your reasoning. You should write **in words** what you are doing and why. The person reading your solution should easily be able to understand what you have done, *because you explained it, in words*. Submissions with little or no written explanations will not receive full marks.
4. **Uploading:** A handwritten assignment can be photographed or scanned. It is important that the images (the scans or photos) are clear and easy to read (not too dark, too bright, too blurry, etc.). Your submitted files are what will be marked, so if the grader(s) can't read them or can't make sense of what you've written, you will not get full marks.

You may also write your solutions on a tablet, or type your solutions (LaTeX only please) as long as your solutions are clear, easy to read and follow, and are all your own work.

**Make sure you upload solutions to each problem to the correct place.** If you upload solutions to the wrong problems, or upload all your solutions to the same problem, it may result in getting 0 on all problems uploaded to the wrong places.

5. **Grading:** The assignment is out of 10 marks, with the marks for each question indicated beside it.  
  
Marks for each solution will depend on your answer, as well as the quality of your explanation of those answers.
6. **Academic Integrity:** The solutions that you submit to this assignment must be all your own work. By submitting this assignment you declare that:
  - (a) Your solutions are all your own work, explained in your own words.
  - (b) You have not copied any part of the assignment solutions from anyone or anywhere.
  - (c) You have not let anyone else copy any part of your solutions to the assignment.
  - (d) You have not used Generative AI (e.g., ChatGPT) for any part of the assignment.

## Background

It's been several weeks since the aliens arrived at UTM. They are usually very friendly, except when they get hungry.

Humanity's biggest challenge has been feeding the aliens. Their supply of native fruit, *gethens*, is running out, and any gethens they expose to our atmosphere ripen extremely quickly and become inedible within seconds.

They are a very sensitive species, and can only eat fruit when it is nearly *perfectly* ripe. You have been tasked with finding the most perfect fruit for Estraven, an alien friend you've made<sup>1</sup>. You will need to carefully analyze how quickly fruit ripens, and find the window of time during which the fruit is acceptably ripe for Estraven.

**Definition:** A **sigmoid** is any function whose domain is  $(-\infty, \infty)$  and whose range is a **subset of**  $(-1, 1)$ <sup>2</sup>.

We will use some sigmoids  $S$  to model the ripeness of a piece of fruit:  $S(t)$  represents the ripeness of that piece of fruit at time  $t$  (measured in days). Our sigmoids will all be normalized so that an output of  $S(t) = 0$  represents the ideal ripeness for Estraven. When  $S(t)$  is near 1, the fruit is completely overripe (and maybe even rotten). When  $S(t)$  is near  $-1$ , the fruit is completely underripe.

Specifically, we'll be interested in the sigmoids like the following (the two expressions are equal):

$$F(t) = \frac{e^{at} - 1}{e^{at} + 1} = 2 \frac{e^{at}}{1 + e^{at}} - 1,$$

where the parameter  $a > 0$  is a positive constant that will differ based on the fruit we're analyzing.<sup>3</sup>

Here's an interactive Desmos graph, to get a sense of what these types of sigmoids look like for different values of  $a$ .

<https://www.desmos.com/calculator/zhbv8dwahx>

You should expect this assignment to require you to spend some time playing with that visualization and thinking about that graph (and tangent lines to that graph) look like. Don't leave it to the last minute!

Finally, in case you haven't talked about it in class just yet, you may need the fact that  $\frac{d}{dx} [e^x] = e^x$ .

## Problems

1. (2 points) Let's do some basic analysis of functions like  $F$ .
  - (a) (0.5 points) In two or three sentences and some algebra, explain why  $F$  is always a sigmoid, no matter what choice of  $a > 0$  we use.
  - (b) (0.5 points) Verify (by computation) that  $F(0) = 0$ , no matter what choice of  $a$  we use. This means that any given fruit is always perfectly ripe at time  $t = 0$ .
  - (c) (1 points) Verify that  $F$  is an odd function, no matter what choice of  $a$  we use.

<sup>1</sup>Estraven is friendly, likes making others laugh, and instinctively trusts strangers. Their close friends call them Estrav!

<sup>2</sup>In real life applications, like the activator functions of neural nets or cumulative distribution in statistics, the range is usually  $(0, 1)$ , and the functions are usually assumed to be differentiable.

<sup>3</sup>If we were being very precise, we would write the function with a subscript like  $F_a$ , to show that every different  $a$  makes it a different function. As long as there's no chance of confusion here, we'll write it without the subscript.

2. (1 point) Our alien friends are very picky about ripeness: they only enjoy fruit that is between  $-0.1$  and  $0.1$  ripeness (inclusive). The aliens will get sick if the ripeness is outside of that range but still between  $-0.5$  and  $0.5$  (inclusive). If the aliens eat a fruit whose ripeness is outside of the range between  $-0.5$  and  $0.5$ , then they will die!

For the sigmoid  $F$  with its general parameter  $a$ , find a time  $t_s > 0$  such that a fruit with ripeness  $F(t_s)$  will make the aliens sick but not kill them, and find a time  $t_d < 0$  such that a fruit with ripeness  $F(t_d)$  will kill them.

*Hint:* Do you expect your answers to depend on  $a$ ?

3. (1 point) Ivan has given you three sigmoids that model fruit: one for apples, one for bananas, and one for gethens. You know the parameters for these three fruit are supposed to be  $a = 0.2$ ,  $a = 1$ , and  $a = 10$ , but you weren't paying attention in tutorial and missed which parameter is supposed to go with which fruit!

Determine which parameter should go with each fruit, and explain in a few sentences why you've chosen the way you did.

4. (1 point) MAT135 students, with their new knowledge of how tangent lines can approximate more complicated functions, are worried about computing exact values of these sigmoids and would prefer to approximate them with tangent lines.

(a) (0.5 points) Find the equations of the tangent lines at  $t = 0$  to the graphs of each of the three sigmoids (for apples, bananas, and gethens) mentioned in Q3.

(b) (0.5 points) Find the equation of the tangent line at  $t = 0$  to the graph of the sigmoid  $F$  for a general parameter  $a$ .

Call this tangent line function  $L$ , so the equation of the tangent line is  $y = L(t)$ . After you find  $L$ , add the equation of the tangent line to the Desmos visualization linked above, and play around with the slider for  $a$  to get a sense of how the graph of  $L$  looks compared to the graph of  $F$ .

5. (2 points) The students are unsure of the accuracy of the tangent line approximation, so they run some tests for bananas and gethens.

(a) (1 point) Using a calculator and the sigmoid model for bananas and gethens, compute the ripeness of a banana and a gethen 6 hours, 12 hours, 1 day, and 3 days after  $t = 0$ . (Remember,  $t$  is in days.)

Then, do the same but using your tangent line model for a banana and a gethen.

Show all your work at least for the gethen computations, round all your answers to three decimal places, and organize all of your final answers clearly in a table like this at the beginning of your submission:

|                                | 6 hours | 12 hours | 1 day | 3 days |
|--------------------------------|---------|----------|-------|--------|
| banana ripeness (sigmoid)      |         |          |       |        |
| banana ripeness (tangent line) |         |          |       |        |
| gethen ripeness (sigmoid)      |         |          |       |        |
| gethen ripeness (tangent line) |         |          |       |        |

(b) (1 point) In at most a few sentences, explain how the accuracy of the tangent line approximations changes depending on  $t$ . When is the tangent line model more accurate or less accurate, and why? For which fruit is the tangent line more accurate, and why?

6. (3 points) Before you use the approximate tangent line models to help you find fruit for Estra, you need to figure out whether using these approximations will endanger them.

- (a) (1 point) Use the sigmoid model to find the precise length of time after  $t = 0$  during which Estra will enjoy eating an apple, and the precise length of time after  $t = 0$  during which Estra will *survive* eating an apple. Then use the tangent line model for an apple to find approximations of the same two times.

Show all your work, and round your answers to three decimal places, if necessary. Clearly state and label the two times you find beside each other in your solution.

- (b) (1 point) Generalize your answer to the previous part by finding these exact lengths of time for the sigmoid model  $F$  and the tangent line model  $L$ , where both have a general parameter  $a$ .

Show all your work, and express your times as exact values in terms of  $a$ . Clearly state and label the two times you find beside each other in your solution.

- (c) (1 point) In a few sentences (and maybe a bit of algebra or computation), explain why using the tangent line models to approximate these times keeps will always keep Estra safe, no matter what fruit they ask for.

Calculus helps humanity avoid an intergalactic war!